WONKYUNG JANG

Assistant Professor
Department of Instructional Leadership and Academic Curriculum
Jeannine Rainbolt College of Education
University of Oklahoma
Phone: (919) 360-3560 | E-mail: wkjang@ou.edu
Faculty Website: https://www.ou.edu/education/people/faculty/wonkyung-jang

EDUCATION

University of North Carolina at Chapel Hill, USA
2022 Ph.D., Education
2021 M.S., Statistics and Operations Research
2021 M.Ed., Early Childhood Intervention and Family Support
Yonsei University, South Korea
2015 B.S., Child and Family Studies
2015 B.A., Applied Statistics
University of Oslo, Norway
2013 Exchange Student, Educational Sciences

PROFESSIONAL EXPERIENCE

2022-Present Assistant Professor (Tenure-Track), Department of Instructional Leadership and Academic Curriculum (ILAC), Jeannine Rainbolt College of Education (JRCoE), University of Oklahoma (OU) (Norman-Based Program; Classified as R1: Doctoral Universities – Very High Research Activity)
2022-Present Principal Investigator, Multimodal Observation of Situated Activity, Interaction, and Child Learning (MOSAIC) Lab [Lab Website: https://wkjang.github.io/]

LICENSURES & CERTIFICATIONS

Graduate Certificate in Computational Linguistics (UNC School of Information and Library Science and Departments of Linguistics and Computer Science) (2021)
Teacher of Kindergarten (Ages 3-5) (Ministry of Education, South Korea) (2015)
Teacher of Child Care Center (Ages 0-5) (Ministry of Health and Welfare, South Korea) (2015)
Sergeant (Republic of Korea Army) (2010)

HONORS & AWARDS

New PI Award (OU Institute for Community and Society Transformation) (2026)
Pre-Tenure Faculty Award – College-Wide (OU JRCoE) (2026)
Pre-Tenure Faculty Award – Department-Level (OU JRCoE ILAC) (2025)
Fellowship for Global Engagement (OU JRCoE ILAC) (2024)
First Place Poster Award (Oklahoma Partnership for School Readiness) (2024)
Vice President for Research and Partnerships (VPRP) Award for Excellence in Transdisciplinary, Convergent Research (OU Office of the VPRP) (2023; as Happy Teacher Project member)
NAECTE Foundation Doctoral Scholarship (National Association of Early Childhood Teacher Educators) (2021)
Tanner Award for Excellence in Undergraduate Teaching (UNC Office of the Provost) (2021)
[Featured] {School of Education} {Department of Statistics and Operations Research}
Best Poster Award (Conference on Artificial Intelligence in Smart Cities) (2021)
ISLS & Wallace Foundation Scholarship (International Society of Learning Sciences) (2021)
Nancy Blanche Norman Scholarship (UNC School of Education) (2021)
Alumni Association in the U.S. Award (Yonsei University College of Human Ecology) (2019)
Graduate Transportation Grant Award (UNC Graduate School) (2017)

GRANTS (Total $700,000 Awarded as PI and Co-PI)

[10] **The JRCoE Faculty Summer Research Grant (2025)**
Proposal: "AI-Supported Professional Learning for Early Scientific Literacies: Strengthening Community-Based Instructional Practice Across Families, Libraries, Museums, and Early Learning Settings" (Principal Investigator (PI), $6,000)
Jeannine Rainbolt College of Education, University of Oklahoma

[9] **Secondary Analyses of Data on Early Care and Education Grant (2023-25)**
Proposal: "Innovative Approaches to Studying Peer Effects in Head Start and Their Implications for Policy, Research, and Practice" (PI, $100,000)
Administration for Children and Families, U.S. Department of Health and Human Services

[8] **Institute for Community and Society Transformation Seed Grant Award (2023-24)**
Proposal: "Leveraging Big Data to Improve Quality in Infants' and Toddlers' Language Learning Environments" (PI, $30,000)
Institute for Community and Society Transformation (ICAST), University of Oklahoma

[7] **Oklahoma Partnership for School Readiness Foundation Grant (2024-25)**
Proposal: "Oklahoma Happy Teacher Project: Strengthening Early Care and Education Workforce" (Co-PI, $100,000)
Oklahoma Partnership for School Readiness

[6] **Oklahoma Teacher Connection Collegiate Grant (2023-27)**
Proposal: "Teacher Bound: A Recruitment Pathway for Oklahoma Early Childhood Education Bachelor Degree Program" (Co-PI, $20,000)
Oklahoma Teacher Connection, Oklahoma State Regents for Higher Education

[5] **Secondary Analyses of Data on Early Care and Education Grant (2022-24)**
Proposal: "Unpacking the Complex Story of Head Start Teacher Turnover: Patterns, Factors, Mechanisms and Outcomes" (Co-PI, $100,000)
Administration for Children and Families, U.S. Department of Health and Human Services

[4] **Institute for Community and Society Transformation Seed Grant Award (2022-23)**
Proposal: "Understanding Equity and Opportunity for Myanmar Refugee Children and their Families" (Co-PI, $30,000)
Institute for Community and Society Transformation (ICAST), University of Oklahoma

[3] **Provost's Undergraduate Research and Creative Activity Summer Fellowship (2024)**
Proposal: "Parental Gender and Contextual Influences on Parent-Child Interactions" (Co-Faculty Mentor, $5,000)
Provost's Undergraduate Research and Creative Activity (UReCA), University of Oklahoma

[2] **George Kaiser Family Foundation Fund (2023-25)**
Proposal: "Tulsa Educare Implementation Study" (Co-PI)
Early Childhood Education Institute (ECEI), University of Oklahoma

[1] **Head Start University Partnership Grant (2024-25)**
Proposal: "Building the Evidence Base for ECE Workforce Well-Being" (Co-I)
Administration for Children and Families, U.S. Department of Health and Human Services

BOOK

Clifford, R., Yazejian, N., **Jang, W.**, & Jigjidsuren, D. (2021). *A guide to analyzing and understanding Early Childhood Environment Rating Scale (ECERS)–3 scores.* Teachers College Press, Columbia University.

PEER-REVIEWED PUBLICATIONS

AI-Enhanced Research on Early Learning and Development

[1] Casey, E. M., **Jang, W.,** Dewhirst, C., Spivey, E., & Zang, H. (2026). The Bandit effect: How caregiver engagement with Bluey influences play attitudes and behaviors. *Early Childhood Education Journal, 54*(6), 5451–5464. https://doi.org/10.1007/s10643-026-02225-5

[2] **Jang, W.**, Horm, D., Kwon, K., Lu, K., Kasak, R., & Park, J. (2025). Leveraging natural

language processing to deepen understanding of parent–child interaction processes and language development. *Family Relations, 74*(3), 1146–1173. https://doi.org/10.1111/fare.13198 [SSCI; Q1]

[3] **Jang, W.**, Ko, K., Ku, S., & Kwon, K. (2025). Family and child characteristics in reading achievement milestones using machine-learning-based survival analysis. *Family Relations*, *74*(3), 1174–1197. https://doi.org/10.1111/fare.13174 [SSCI; Q1]

[4] **Jang, W.**, Kwon, K., & Horm, D. (2024). The role of language and literacy skills in science learning from kindergarten to 5th grade: Mitigating gender, racial/ethnic, and socio-economic disparities. *Education Sciences, 14*(9), 994. https://doi.org/10.3390/educsci14090994 [ESCI; Q1]

[5] **Jang, W.** (2024). Bridging the science achievement gap: Using survival analysis to assess the impact of center-based early childhood programs on children with disabilities. *Journal of Infrastructure, Policy and Development, 8*(16), 6776. https://doi.org/10.24294/jipd6776

[6] Lake, V., Ghorbani, E., Lim, B.Y., Ruan, J., Noyori-Corebett, C., Abbas, J., & **Jang, W.** (2024). Tech challenges & opportunities: Myanmar refugee families in the U.S. *International Journal of Early Childhood Education, 30*(2), 153–172. https://doi.org/10.18023/ijece.2024.30.2.007 [KCI]

[7] **Jang, W.**, & Leech, K. (2023). Contextual modulation of adult-child language interaction: Semantic network connectivity and children's vocabulary development. *Education Sciences, 13*(11), 1084. https://doi.org/10.3390/educsci13111084 [ESCI; Q1]

[8] **Jang, W.** (2022). *Early childhood teachers' characteristics, complex talk, and children's language and literacy development* [Doctoral dissertation, The University of North Carolina at Chapel Hill]. ProQuest Dissertations and Theses Global. https://doi.org/10.17615/00ww-d094 (Advisors: Drs. Margaret Burchinal & Harriet Able)

[9] Cohen-Vogel, L., Little, M., **Jang, W.**, Burchinal, M., & Bratsch-Hines, M. (2021). A missed opportunity? Instructional content redundancy in pre-k and kindergarten. *AERA Open, 7*(1), 1-15. https://doi.org/10.1177/23328584211006163 [SSCI; Q1]

[10] **Jang, W.**, Francisco, J., Ranganathan, N., McCarroll, K. M., & Ryoo, K. (2020). Using machine learning to understand students' learning patterns in simulations. In Gresalfi, M. and Horn, I. S. (Eds.), *The Interdisciplinarity Society of the Learning Sciences, 14th International Conference of the Learning Sciences (ICLS) 2020, Volume 5* (pp. 2593-2596). Nashville, TN: International Society of the Learning Sciences. https://repository.isls.org//handle/1/6627 [SCOPUS]

[11] Kwon, K., **Jang, W.**, Ford, T. G., Horm, D., & Yazejian, N. (2026). Within-year teacher turnover and children's developmental outcomes in Early Head Start and Head Start programs in the Post-COVID-19 Era. *Early Childhood Research Quarterly, 76*(3), 101–112. https://doi.org/10.1016/j.ecresq.2026.01.006 [SSCI; Q1]

Causal Analytics for Early Childhood Research and Policy

[1] Kwon, K., Ford, T. G., Cheema, C., Kile, M., Wang, H., Tsotsoros, J., Randall, K, Lloyd-Jones, B., Jang, S., Jung, Y., Lee, H., Kim, S., Sorrels, B., Wang, L., Horm, D., & **Jang, W.** (2026). Head Start staff well-being: Early impact results from a cluster randomized control trial (RCT) of a Holistic Wellness Intervention. *Early Childhood Research Quarterly, 76*(3), 74–86. https://doi.org/10.1016/j.ecresq.2026.01.001 [SSCI; Q1]

[2] Kwon, K., **Jang, W.**, Ford, T. G., & Castle, S. (2025). The Head Start teacher paradox: Working conditions, whole well-being, and classroom quality. *Early Childhood Research Quarterly, 72*, 371–382. https://doi.org/10.1016/j.ecresq.2025.04.003 [SSCI; Q1]

[3] Kwon, K., **Jang, W.**, Ford, T. G., Horm, D., Yazejian, N., & Bryant, D. (2025). A longitudinal study of Head Start teacher turnover trends and factors. *Teaching and Teacher Education, 156*, 104916. https://10.1016/j.tate.2024.104916 [SSCI; Q1]

[4] Min, J., Cheng, W., Kim, S. G., Kwon, K., Ford, T. G., & **Jang, W.** (2025). Early childhood

teachers' job demands and relationships with children: The moderating role of work climate. *Early Childhood Education Journal, 54*(3), 1755-1765. https://doi.org/10.1007/s10643-025-01957-0 [SSCI; Q2]

[5] Bryant, D., Yazejian, N., **Jang, W.**, Kuhn, L., Hirschstein, M., Soliday Hong, S., Stein, A., & Educare Learning Network Investigators. (2023). Retention and turnover of teaching staff in a high-quality early childhood network. *Early Childhood Research Quarterly, 65*, 159–169. https://doi.org/10.1016/j.ecresq.2023.06.002 [SSCI; Q1]

Technical Report

[1] Kwon, K., Ford, T. G., & **Jang, W.** (2025). Oklahoma early childhood workforce study report. Oklahoma Partnership for School Readiness. With contributions from Ambrianna Freeze, Sun Geun Kim, Junghyun Min, Catlin Carter, Huiying Zhang, Weiyi Cheng, Yuenjung Joo, & Raissa Alana Pereiral.

UNDER REVIEW | REVISE & RESUBMIT

[6] **Jang, W.** From prediction to prevention: Human-in-the-loop AI for strengthening quality of care in child- and family-serving systems.

[5] **Jang, W.** Responsible AI for child and family well-being: A human-centered framework for strengthening relationships, services, and communities.

[4] **Jang, W.**, Horm, D., Castle, S., Park, Y., Acton, M., & Jackson, B. Examining the psychometric properties of the Infant and Toddler Literacy Assessment (ITLA).

[3] **Jang, W.** Using survival analysis to study the timing of children surpassing a developmental threshold.

[2] Park, Y., **Jang, W.**, Horm, D., Acton, M., & Gallagher, K. Exploring the mechanisms linking parent-teacher relationships to child outcomes.

[1] Kile, M., Kwon, K., Wang, H., Kim, S., Min, J., **Jang, W.**, Cheng, W., & Rasby, S.O. The importance of the physical environment to support well-being of children and teachers in early care and education settings.

MANUSCRIPTS IN PREPARATION

[10] J**ang, W.** AI, digital equity, and childhood well-being: Understanding linguistic, cultural, developmental, and socioeconomic disparities in emerging technologies.

[9] **Jang, W.** PLACE-AI: Examining how generative AI represents place, culture, community knowledge, and belonging in child- and family-serving contexts.

[8] **Jang, W.,** & Educare Learning Network Team. Relational continuity as protective infrastructure for childhood well-being: A computational social network approach to families, professionals, and systems of care.

[7] **Jang, W.** Multilingual family-provider communication in the age of AI: Language, belonging, digital equity, and quality of care.

[6] **Jang, W.** Gendered language environments in early childhood: Using computational linguistics to examine identity, relationships, and developmental well-being.

[5] **Jang, W.**, & Jun, H. Cultural models of father involvement in Norway and South Korea: Integrating deep learning, family relationships, culture, and child well-being.

[4] **Jang, W.** Semantic network development among typical and late talkers: A longitudinal computational approach to communication, social participation, and developmental well-being.

[3] **Jang, W.**, Burchinal, M., & Early Learning Network Team. Language-rich environments across children's everyday contexts: Activity settings, relational experiences, and language and literacy development.

[2] **Jang, W.**, Burchinal, M., & Early Learning Network Team. Caregiver characteristics and complex talk: Relational pathways to children's language, literacy, and developmental well-being.

[1] **Jang, W.**, Kwon, K., Horm, D., Ford, T., & Yazejian, N. Peer relationships and social networks as developmental contexts for language, literacy, belonging, and well-being in early childhood.

UNIVERSITY TEACHING EXPERIENCE

Assistant Professor, OU Jeannine Rainbolt College of Education (2-2 Teaching Load)
ILAC 6043: Multilevel Modeling (2023, 2026)
ILAC 6413: Advanced Data Science Methods in Social Sciences (2026)
ILAC 6930: Advanced Quantitative Research (2023)
EDEC 5533: Advanced Studies in Infant-Toddler Development (2025)
EDEC 6303: Parent Involvement and Education (2024)
EDEC 3513: Integrated Curriculum (Birth Through 5) (2023, 2024, 2025, 2026)
EDEC 3523: Field Experience with Seminar (2023, 2024, 2025, 2026)
EDEC 3533: Infant/Toddler Development and Care (2022, 2023, 2024, 2025)
Teaching Fellow (Instructor), UNC Department of Statistics and Operations Research
STOR 155: Introduction to Data Models and Inference (Summer 2020, Summer 2021)
STOR 320: Introduction to Data Science (Summer 2021)
Teaching Assistant, UNC School of Education
EDUC 181: Introduction to Human Development and Family Studies (Fall 2018, Spring 2019)
EDUC 405: Parenting and Family Life Education (Fall 2020, Spring 2021)
EDUC 408: Research Methods in Human Development (Fall 2019, 2020, Spring 2020, 2021)
EDUC 601: Education Workshop (Summer 2021)
EDMX 767: Infant/Toddler Curriculum and Learning Environments (Summer 2021; Co-instruct)
Teaching Fellow (Instructor), UNC Department of Asian Studies
KOR 102: Elementary Korean 2 (Spring 2017)
Teaching Associate, UNC Department of Psychology and Neuroscience
PSYC 210: Statistical Principles of Psychological Research (Fall 2017, Spring 2018)
Guest Lecturer, UNC, PSYC 250: Child Development (Summer 2019, Fall 2019); KOR 101: Elementary Korean 1 (Fall 2017); KOR 203: Intermediate Korean 1 (Fall 2017)

RESEARCH SKILLS

Statistics and Computer Science Coursework (83 credits completed)
Undergraduate-Level Courses from Yonsei University
Calculus, Linear Algebra, Mathematical Statistics, Introduction to Statistics, Statistical Method, Sampling Theory, Regression Analysis, Experimental Design, Nonparametric Statistics, Time Series Analysis, Computer Programming, Data Analysis and Design, Practical Risk Management and Statistics
Graduate-Level Courses from UNC
Mathematical Statistics, Applied Statistics 1, Introduction to Data Science, Machine Learning, Data Mining, Object Oriented Data Analysis, High-Dimensional Probability with Applications in Statistics and Machine Learning, Text Mining, Extreme Value Theory, Multilevel Modeling, Introduction to Structural Equation Modeling, Statistical Analysis of Educational Data, Secondary Data Analysis, Masters Essay
Other Statistics Trainings
Causal Inference, Bayesian Statistics, Item Response Theory, Dyadic Data Analysis, Missing Data Treatment, Meta-analysis, Scale Development
Data Science & Programming Languages
Quantitative Analysis Softwares
R, Python, SAS, Mplus, STATA, MINITAB, SPSS, AMOS, LISREL, JAVA, MS-SQL
Tools for Corpus Linguistics/Multi-Modal Data Analysis
Python, R, CLAN, ELAN, PRAAT, NVIDIA Toolkit
Eye-Tracking System

Python, Tobii Pro Lab, Yarbus, GazeTag
Qualitative Analysis Softwares
MAXQDA, ATLAS.ti, NVivo

SELECTED PRESENTATIONS

[57] **Jang, W.**, Ssentuuwa, R., Abeyrathna, D., & Lake, V. (2026, June). Preparing teachers for equitable science learning: Integrating language, literacy, and social-emotional development. Paper presented at the 2026 National Association of Early Childhood Teacher Educators (NAECTE) Summer Conference.

[56] Kuhn, L., **Jang, W.**, Park, Y. R., Figuero, C., Dean, M., Telfer, N., & Faria, A.-M. (2026, June). Turnover trends in early care and education family support staff. Paper to be presented at the 2026 National Research Conference on Early Childhood Conference (NRCEC) Conference.

[55] Kwon, K., **Jang, W.**, Ford, T. G., Horm, D., & Yazejian, N. M. (2026, June). Teacher turnover and children's developmental outcomes in Early Head Start and Head Start programs in the post-COVID-19 era. Poster to be presented at the 2026 National Research Conference on Early Childhood (NRCEC) Conference.

[54] Gallagher, K., & **Jang, W.** (2026, June). Early Head Start classroom language support and child development in the transition to preschool. Poster to be presented at the 2026 National Research Conference on Early Childhood (NRCEC) Conference.

[53] **Jang, W.**, Gallagher, K., & Kelly, C. (2026, February). Raising children together: Building inclusive family-teacher relationships in Head Start settings. Poster to be presented at the 2026 Conference on Research Innovations in Early Intervention (CRIEI).

[52] **Jang, W.** (2025, December). From relationships to growth: Integrating Machine Learning and Social Network Analysis to link Head Start network dynamics with children's learning outcomes. Paper presented at the 2025 Educare Data Camp, Chapel Hill, NC.

[51] Gallagher, K., **Jang, W.**, Kelly, C., Acton, M., Rives, B., & Carolina, A. (2025, May). Raising children together: Parent–teacher relationships and children's development. Paper presented at the 2025 Educare Network Meeting, Flint, MI.

[50] Kuhn, L., & **Jang, W.** (2025, May). Examining factors that contribute to turnover of Educare family engagement staff. Paper presented at the 2025 Educare Network Meeting, Flint, MI.

[49] **Jang, W.**, Kwon, K., Horm, D., Ford, T. G., & Yazejian, N. (2025, April). Innovative approaches to studying peer effects in Head Start. Paper presented at the 2025 Biennial Meeting of the Society for Research in Child Development (SRCD), Minneapolis, MN.

[48] Kwon, K., Ford, T. G., & **Jang, W.** (2025, April). Exploring the relationship between working conditions and whole teacher well-being: A latent profile analysis. Paper presented at the 2025 Biennial Meeting of the Society for Research in Child Development (SRCD), Minneapolis, MN.

[47] Kwon, K., Ford, T. G., & **Jang, W.** (2025, April). Holistic approaches to key issues of the early childhood workforce: Working conditions, well-being, and retention. Paper presented at the 2025 Annual Meeting of the American Educational Research Association (AERA), Denver, CO.

[46] Kwon, K., Isomya, L., Min, J., Cheng, W., Zhang, H., Raadschelders, K., Story, C., Castilglione, J., Abbott, K., Ford, T. G., **Jang, W.** (2025, March). Thriving together: Evidence-based strategies for retention and well-being in the early childhood workforce. Paper presented at the 2025 OU KEY Conference, Norman, OK.

[45] Kuhn, L., & **Jang, W.** (2024, December). Understanding Educare family engagement staff turnover: Patterns and factors. Paper presented at the 2024 Educare Data Camp, Chapel Hill, NC.

[44] Park, Y., Horm, D., **Jang, W.**, & Acton, M. (2024, November). Profiles of family stress and

support and links to parent-child relationships and activities. Paper presented at the 2024 National Council on Family Relations (NCFR) Conference, Bellevue, WA.

[43] Kim, K., **Jang, W.**, Kwon, K., Jeon, H., & Min, J. (2024, October). Text mining mothers' and fathers' language during toddler free play for sentimental associations. Poster presented at the 2024 OU UReCA Showcase, Norman, OK.

[42] **Jang, W.**, Horm, D., Park, Y., & Acton, M. (2024, September). The role of cocaring relationships and classroom quality in Head Start programs: Enhancing children's language and literacy. Poster Presented at the 2024 Oklahoma Early Childhood Research Symposium, Oklahoma City, OK.

[41] **Jang, W.**, Park, Y., Horm, D., & Acton, M. (2024, September). Attendance in an enhanced early head start/head start program in Oklahoma: Importance, challenges, and strategies. Poster Presented at the 2024 Oklahoma Early Childhood Research Symposium, Oklahoma City, OK.

[40] **Jang, W.**, Kwon, K., & Ford, T. G. (2024, September). Exploring the relationship between working conditions and teacher well-being: A latent profile analysis. Poster presented at the 2024 Oklahoma Early Childhood Research Symposium, Oklahoma City, OK. **[Best Poster Award]**

[39] Min, J., Cheng, W., Kim, S. G., Kwon, K., Ford, T. G., & **Jang, W.** (2024, September). Association between early childhood teachers' job demands and relationship with children: The moderating role of work climate. Poster presented at the 2024 Oklahoma Early Childhood Research Symposium, Oklahoma City, OK.

[38] **Jang, W.**, Horm, D., Park, Y., & Acton, M. (2024, June). The role of cocaring: Enhancing the effect of classroom quality on children's language and literacy skills. Paper presented at the 2024 National Research Conference on Early Childhood (NRCEC) Meeting, Washington D.C.

[37] **Jang, W.**, Kwon, K., Horm, D., Ford, T. G., & Yazejian, N. (2024, June). Revealing dynamics: Investigating the influence of classroom heterogeneity and prolonged peer exposures on the intensity and nature of peer effects in Head Start classrooms. Poster presented at the 2024 National Research Conference on Early Childhood (NRCEC) Meeting, Washington D.C.

[36] **Jang, W.**, Kwon, K., Horm, D., Ssentuuwa, R., & Abeyrathna, D. (2024, June). Utilizing survival analysis to examine the timing of children surpassing developmental threshold: Mitigating disparities in science achievement. Poster to be presented at the 2024 National Research Conference on Early Childhood (NRCEC) Meeting, Washington D.C.

[35] Min, J., Cheng, W., Kim, S., Kwon, K., Ford, T. G., & **Jang, W.** (2024, June) Association between early childhood teachers' job demands and relationship with children: The moderating role of work climate. Poster presented at the 2024 National Research Conference on Early Childhood (NRCEC) Meeting, Washington D.C.

[34] Kwon, K., **Jang, W.**, Ford, T. G., & Castle, S. (2024, June). The Head Start teacher paradox: Working conditions, well-being, and classroom quality. Paper presented at the 2024 National Research Conference on Early Childhood (NRCEC) Meeting, Washington D.C.

[33] Kwon, K., **Jang, W.**, Horm, D., Ford, T. G., Yazejian, N., & Malek-Lasater, A. (2024, June). Infant-toddler teacher workforce stability–patterns, factors, and outcomes. Paper presented at the 2024 National Research Conference on Early Childhood (NRCEC) Meeting, Washington D.C.

[32] **Jang, W.**, Park, Y., Horm, D., Acton, M., Weeden-Umelo, S., & Morris, M. (2024, May). Attendance at Tulsa Educare: Importance, challenges, and strategies. Paper presented at the 2024 Educare Learning Network Meeting. Sioux City, IA.

[31] Kwon, K., **Jang, W.**, Ford, T., Horm, D., & Yazejian, N. (2024, May). Unpacking the complex story of Educare teacher turnover: Patterns, factors, mechanisms, and outcomes. Paper presented at the 2024 Educare Learning Network Meeting. Sioux City, IA.

[30] Ssentuuwa, R., **Jang, W.**, Horm, D., & Acton, M. (2024, March). The social-emotional development of foster and non-foster children in early care and education settings: The role of dual language learning. Lightning talk at the 2024 OU JRCoE Graduate Student Symposium, Norman, OK.

[29] Jin, L., Ruan, J., & **Jang, W.** (2023, November). Relationships among reading motivation, gender, and reading achievement of Chinese students transitioning from 4th to 6th grade. Paper presented at the 2024 Literacy Research Association (LRA) Conference, Atlanta, GA.

[28] Kwon, K., **Jang, W.**, Ford, T., Horm, D., & Yazejian, N. (2023, June). Longitudinal Head Start teacher turnover study. Paper presented at the 2023 Child Care and Early Education Policy Research Consortium (CCEEPRC) Meeting, Washington D.C.

[27] **Jang, W.** (2023, June). Early childhood teachers' characteristics, complex talk, and children's language and literacy development. Paper presented at the 2023 National Association of Early Childhood Teacher Educators (NAECTE) Conference. Virtual.

[26] Clifford, R., Yazejian, N., **Jang, W.**, & Jigjidsuren, D. (2023, April). A guide to analyzing and understanding ECERS-3 scores with authors. Environment Rating Scales (ERS) National Conference, Alexandria, VA.

[25] **Jang, W.**, & Lim, B. (2023, March). Using distributional semantics to understand preschoolers' depth of vocabulary. Poster presented at the 2023 Society for Research in Child Development (SRCD) Conference, Salt Lake City, UT.

[24] Kwon, K., **Jang, W.**, Kim, S., Jenson, F., Li, Hong., & Zulkifli, A. (2023, March). A holistic view of factors for early care and education (ECE) teacher turnover intent. Poster presented at the 2023 Society for Research in Child Development (SRCD) Conference, Salt Lake City, UT.

[23] **Jang, W.**, & Ko, K. (2022, November). Application of regularized regression in child care quality research: Overcoming the reproducibility crisis. Paper presented at the 84th Annual National Council on Family Relations (NCFR) Conference, Minneapolis, MN.

[22] **Jang, W.**, & Leech, K. (2022, April). Contextual modulation of parent-child conversation: Semantic network connectivity and children's vocabulary development. Poster presented at the 2022 Cognitive Development Society (CDS) Conference, Madison, WI.

[21] **Jang, W.**, & Burchinal, M. (2022, April). Complex preschool teacher talk, content area, and child outcome. Paper presented at the 2022 American Educational Research Association (AERA) Conference, San Diego, CA.

[20] **Jang, W.** (2021, November). The contextual modulation of academic language in early care and education: Implications for B-K teacher education. Paper presented at the 2021 National Association of Early Childhood Teacher Educators (NAECTE) Conference. Virtual.

[19] **Jang, W.** (2021, November). Using network science as a tool for understanding adult-child language interaction in diverse activity contexts. Paper presented at the 83rd Annual National Council on Family Relations (NCFR) Conference. Virtual.

[18] **Jang, W.** (2021, September). Applications of computational grounded theories in inquiry-based learning research. Paper presented at the 30th Annual European Early Childhood Education Research Association (EECERA) Conference. Virtual.

[17] **Jang, W.** (2021, June). Cultural models of paternal involvement in early childhood education: Implications for preservice and inservice teacher education. Paper presented at the 2021 National Association of Early Childhood Teacher Educators (NAECTE) Conference. Virtual.

[16] **Jang, W.** (2021, April). Applications of natural language processing (NLP) in educational research. Poster presented at the 2021 Conference on Artificial Intelligence in Smart Cities. Virtual. **[Best Poster Award]**

[15] **Jang, W.**, Yazejian, N., & Clifford, R. (2021, April). Characteristics of programs, teachers and classrooms: Do features mandated by Head Start Act predict classroom quality? Paper

presented at the 2021 Early Childhood Environment Rating Scale (ECERS) International Conference. Virtual.

[14] **Jang, W.**, Garber, K., Foster, T., Burchinal, M., Bratsch-Hines, M., & Franco, X. (2021, April). Complex preschool teacher-child talk, types of activity settings, and child outcomes. Poster presented at the 2021 Society for Research in Child Development (SRCD) Conference. Virtual.

[13] Foster, T., Garber, K., **Jang, W.**, Burchinal, M., & Bratsch-Hines, M. (2021, April). Examining the role of peer effects for dual language learners in pre-kindergarten. Individual Flash Talk at 2021 Society for Research in Child Development (SRCD) Conference. Virtual.

[12] **Jang, W.** (2020, November). Word embeddings and their relations to distributional semantics in educational linguistics research. Paper presented at the 16th UNCG Regional Mathematics and Statistics Conference (RMSC). Virtual.

[11] **Jang, W.** (2020, November). Inquiry-based learning in early childhood education: Automated text mining analysis of three decades of research literature. Paper presented at the 2020 National Association of Early Childhood Teacher Educators (NAECTE) Conference. Virtual.

[10] **Jang, W.** (2020, June). Developing early childhood teachers' literacy pedagogical content knowledge. Paper presented at the 2020 National Association of Early Childhood Teacher Educators (NAECTE) Conference. Virtual.

[9] **Jang, W.**, Francisco, J., Ranganathan, N., McCarroll, K. M., & Ryoo, K. (2020, June). Using machine learning to understand students' learning patterns in simulations. Paper presented at the 2020 Biennial International Conference of the Learning Sciences (ICLS) Conference. Virtual.

[8] **Jang, W.**, & Wolczynski, N. (2019, November). When early childhood education meets machine learning: Issues with reproducibility and replicability. Paper presented at the 15th UNCG Regional Mathematics and Statistics Conference (RMSC), Greensboro, NC.

[7] Frederick, H., & **Jang, W.** (2019, November). Teaching human development and family studies through case studies: A multi-cultural and problem-based learning approach. Roundtable Presentation at the 81st Annual National Council on Family Relations (NCFR) Conference, Fort Worth, TX.

[6] **Jang, W.** (2019, April). Promoting literacy in early childhood education: Tension between psycholinguistics and sociolinguistics. Round Table Session at the 2019 Korean American Educational Research Association (KAERA) Annual Conference, Toronto, Canada.

[5] **Jang, W.** (2019, April). Aesthetic dimensions of democracy: Analyzing Deweyan pragmatism and Korean democratic education using text mining. Round Table Session at the 2019 American Educational Research Association (AERA) Annual Conference, Toronto, Canada.

[4] **Jang, W.** (2019, March). Early literacy teacher education: Broadening the definitions of pedagogical content knowledge. Paper presented at the 2019 Southeastern Association of Educational Studies (SEAES) Conference, Chapel Hill, NC.

[3] **Jang, W.** (2018, March). Power relations in scientific inquiry during dialogic reading with preschoolers. Paper presented at the 2018 UNC School of Education Graduate Research Symposium, Chapel Hill, NC.

[2] **Jang, W.**, & Jun, H. (2017, November). Scandi daddy and Confucian father in early child education and care: Father involvement in Norway and South Korea from cross-cultural perspectives. Paper presented at the 79th Annual National Council of Family Relations (NCFR) conference, Orlando, FL.

[1] **Jang, W.**, & Jun, H. (2017, August). Cross-cultural perspectives on paternal involvement in early childhood education. Paper presented at the 27th Annual European Early Childhood Education Research Association (EECERA) conference, Bologna, Italy.

WEBINARS AND WORKSHOPS

[9] **Jang, W.** (2025, March). Transforming early care and education with AI and Machine Learning: Using Big Data to improve quality in early learning environments. SKKU-OU Global Dialogues in Early Childhood Education, Seoul.

[8] **Jang, W.** (2025, February). Reimagining society through Artificial Intelligence and Machine Learning: The power of Big Data in child and family studies. Seoul National University, Virtual.

[7] **Jang, W.,** & Park, S. (2024, December). Leveraging Machine Learning and Natural Language Processing algorithms for a deeper understanding of adult-child language interactions. University of Hong Kong, Virtual.

[6] **Jang, W.** (2024, March). Introduction to Machine Learning. National Council on Family Relations, Virtual.

[5] **Jang, W.** (2023, November). Leveraging Big Data to improve the quality in language and literacy learning environments. Ewha Womans University, Virtual.

[4] **Jang, W.** (2023, August). Application of Machine Learning and Natural Language Processing in child and family studies. Seoul National University, Virtual.

[3] Clifford, R., Yazejian, N., **Jang, W.**, & Jigjidsuren, D. (2022, September). A guide to analyzing and understanding ECERS-3 scores. Teachers College Press Webinar, Virtual.

[2] **Jang, W.** (2022, August). The 4th industrial revolution and its impact on early childhood education. National Council on Family Relations (NCFR) Korean Families Focus Group, Virtual.

[1] **Jang, W.** (2022, August). How Machine Learning and AI are transforming the early care and education sector. Sungkyunkwan University Social Innovation Convergence Program, Virtual.

DOCTORAL & THESIS COMMITTEES

[1] Chair, Doctoral Committee, Ronald Ssentuuwa, Department of Instructional Leadership and Academic Curriculum, Jeannine Rainbolt College of Education, University of Oklahoma, Norman.

[2] Member, Doctoral Dissertation Committee, Sun Geun Kim, Triadic dual language storybook reading for adjustment of Afghan refugee children, Department of Instructional Leadership and Academic Curriculum, Jeannine Rainbolt College of Education, University of Oklahoma, Norman.

[3] Member, Doctoral Dissertation Committee, Graydon Morris, Parents' perceptions of debated educational terms and issues based on political orientation: A qualitative inquiry. Department of Educational Leadership and Policy Studies, Jeannine Rainbolt College of Education, University of Oklahoma, Norman.

[4] Member, Doctoral Dissertation Committee, Boo-Young Lim, Supporting dual language learners in Head Start: Peers and teachers' language context, Department of Instructional Leadership and Academic Curriculum, Jeannine Rainbolt College of Education, University of Oklahoma, Norman.

[5] Member, Doctoral Dissertation Committee, Michael Wheelus, Can self-organizing teacher groups reduce teacher isolation? Department of Educational Leadership and Policy Studies, Jeannine Rainbolt College of Education, University of Oklahoma, Norman.

[6] Member, Doctoral Committee, Jenny Min, Department of Instructional Leadership and Academic Curriculum, Jeannine Rainbolt College of Education, University of Oklahoma, Norman.

[7] Member, Doctoral Committee, Emily Cheng, Department of Instructional Leadership and Academic Curriculum, Jeannine Rainbolt College of Education, University of Oklahoma, Norman.

[8] Member, Doctoral Committee, Ronnie Titsworth, Department of Educational Leadership and Policy Studies, Jeannine Rainbolt College of Education, University of Oklahoma, Norman.
[9] Member, Doctoral Committee, Kim Bastin, Department of Educational Leadership and Policy Studies, Jeannine Rainbolt College of Education, University of Oklahoma, Norman.
[10] Member, Doctoral Committee, Justin Trujillo, Department of Educational Leadership and Policy Studies, Jeannine Rainbolt College of Education, University of Oklahoma, Norman.

REVIEWER FOR JOURNAL

Asia Pacific Education Review
Child Development
Early Childhood Research Quarterly
Quality & Quantity

REVIEWER FOR CONFERENCE

National Association of Early Childhood Teacher Educators (NAECTE)
National Council on Family Relations (NCFR)
International Society of the Learning Sciences (ISLS)
Society for Research in Child Development (SRCD)

SERVICE ---------- STATE/NATIONAL/INTERNATIONAL

Oklahoma State Regents for Higher Education

Course Equivalency Project (CEP) Curriculum Committee (Child Development) (2025)

National Council on Family Relations Korean Families Focus Group

NCFR Korean Families Focus Group Mentor-Mentee Program (2023)

Korean-American Educational Researchers Association

Invited Talk

[3] **Jang, W.** (2023, September). Exploring opportunities: Job market research presentation. Korean-American Educational Researchers Association (KAERA), Virtual.
[2] **Jang, W.** (2023, August). Preparing for the academic job market. Korean-American Educational Researchers Association (KAERA), Virtual.
[1] **Jang, W.** (2022, March). Preparing for the academic job market. Korean-American Educational Researchers Association (KAERA), Virtual.

SERVICE ---------- UNIVERSITY/COLLEGE/DEPARTMENT/PROGRAM

University of Oklahoma

OU JRCoE Search Committee Member: Early Reading/Literacy Faculty Search (2025-2026)
OU JRCoE MA in Childhood Well-Being Program Coordinator (2024-2026)
OU JRCoE Teacher Bound (2023-2026)
OU JRCoE Search Committee Member: GKFF Endowed Chair Search (2023-2024)
OU-Tulsa Research Symposium Judge (2023-2025)
OU JRCoE Education Profession Division Council (2022-2024)

Invited Talk

[2] **Jang, W.** (2025, October). Transforming ECE with multimodal AI: Harnessing cutting-edge Artificial Intelligence and Machine Learning to advance teaching, learning, and quality in early learning environments. OU JRCoE Center for Educational Development and Research, Norman, OK.
[1] **Jang, W.** (2022, October). When Machine Learning meets early childhood education: Leveraging Big Data to improve quality in language learning environments. OU Dodge College of Arts and Sciences, Norman, OK.

University of North Carolina at Chapel Hill

UNC School of Education GSA Diversity and Equity Chair (09.2021-08.2022)
UNC University Teaching Awards Committee (07.2021-01.2022)

UNC & Duke Wonder of Learning Exhibition on Reggio Emilia (02.2017-04.2017)

Invited Talk

[7] Jones, K., & **Jang, W.** (2024, December). Optimizing data management strategies for Educare. FPG Child Development Institute, Chapel Hill, NC.

[4] **Jang, W.**, & Kuhn, L. (2021, September). The influence of personal and program characteristics on teacher retention in Educare: Using multiplicative and additive hazards regression models. FPG Child Development Institute, Chapel Hill, NC.

[3] **Jang, W.** (2021, March). Parent-child interaction corpus analysis: Using computational linguistics and machine learning approaches. FPG Child Development Institute Brain and Early Experiences Lab, Chapel Hill, NC.

[2] **Jang, W.** (2021, February). Applications of hierarchical Bayesian modeling in early childhood research. FPG Child Development Institute Data Management and Analysis Core, Chapel Hill, NC.

[1] **Jang, W.** (2019, November). How machine learning algorithms can be used to help mine hidden trends and patterns: Using R for visualization and data mining. FPG Child Development Institute Educare Learning Network Datacamp, Chapel Hill, NC.

TEACHING EXPERIENCE IN EARLY CHILDHOOD SETTINGS

Teacher

Banpo Haneul Preschool (Full-time; 03.2016-07.2016); Chapel Hill-Durham Korean School (Part-time; 08.2017-05.2022)

Volunteer

Catholic Dominic Order (07.2007-08.2007); Yonsei Province Child and Adolescent Center (09.2007-12.2007); Marista Community Center (09.2007-12.2007); Child Fund Korea (07.2011-08.2011); Yonsei University Child Development Research Institute (04.2013-05.2013; 07.2011); Puruni Seocho Preschool (Student Teaching Internship; 07.2013-08.2013); Youngwon Middle School (03.2014-06.2014); Hanyoung Kindergarten (Student Teaching Internship; 03.2014); Yonsei University Rehabilitation School (09.2014-12.2014); Kindergarten of Chongshin University (03.2015-06.2015)

PROFESSIONAL AFFILIATIONS

American Educational Research Association (AERA)
European Early Childhood Education Research Association (EECERA)
International Society of the Learning Sciences (ISLS)
Literacy Research Association (LRA)
National Association of Early Childhood Teacher Educators (NAECTE)
National Association for the Education of Young Children (NAEYC)
National Council of Family Relations (NCFR) (& Korean Families Focus Group)
Society for Research in Child Development (SRCD) (& Asian Caucus Group)